

# From Games to Graphs: A Categorical Framework for Evolutionary Diversity

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## Abstract

We present a unified categorical framework connecting combinatorial game theory, graph theory, and evolutionary computation. The key insight is that migration topologies in genetic algorithms, game trees in combinatorial games, and spectral properties of graphs all arise as instances of a single categorical construction involving enriched profunctors. We formalize this using the theory of enriched categories and demonstrate practical applications to algorithm design.